

Ad Safety Moderation Pipeline

Computer vision capstone report

MS_DSP_462 | August 2026

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1. PROBLEM AND OBJECTIVES

We built a local prototype that reviews one static ad at a time and returns **APPROVE**, **REVIEW**, or **BLOCK**, with the evidence behind the result. It helps sort a review queue; it does not decide legality, audience eligibility, landing-page claims, or jurisdictional compliance [1, 2].

- Classify four in-scope image categories on a grouped final test split.
- Add optional object and OCR evidence without letting those signals block an ad by themselves.
- Show scores, reasons, timing, overlays, and a downloadable audit record for the reviewer.

2. DATASET AND PREPROCESSING

The formal registry contains **288** images in **215** source or campaign groups. Grouped splitting produced train/validation/test counts of 192/48/48. Verified overlap across splits: 0 source groups and 0 exact hashes.

Label	Train	Val	Test	Total
Safe / general advertising	48	12	12	72
Firearms and ammunition	48	12	12	72
Explosives and explosive devices	48	12	12	72
Financial or cryptocurrency promotion	48	12	12	72

Registry-observed dimensions span 253-768 px wide and 111-768 px high. Encodings: image/jpeg: 288. Synthetic-declared rows: 144/288. External sources: rajshivanshu/weapons_set1 (144), EthanGabis/ADautoGen-DS (72) [7, 8]. We used a Pillow-based project script to create the remaining 72 synthetic finance creatives.

- Source or generated campaign groups stay inside one split; exact-file and group overlap counts are both zero.
- RGB input is resized to 248 px, center-cropped to 224 by 224 with bicubic interpolation, then normalized with mean and standard deviation (0.5, 0.5, 0.5).
- The pinned timm transform runs before frozen feature extraction; StandardScaler learns from training embeddings only.

3. ARCHITECTURE AND DECISION EQUATIONS

A pinned ViT-B/16 backbone (85,798,656 parameters, all frozen) converts each RGB creative into a 768-dimensional embedding. A StandardScaler and multinomial logistic head produce four class probabilities [3, 5].

Grounding DINO and Tesseract are optional evidence channels. The policy layer applies noisy-OR fusion to restricted cues; only the calibrated classifier can trigger an automatic firearms or explosives block [4, 9].

$$z = f_A(x) \in \mathbb{R}^{768}; u = (z - \mu) / \sigma$$

$$P(y = k | x) = \exp(w_k^T u + b_k) / \sum_i \exp(w_i^T u + b_i)$$

$$\text{fused score}_k = 1 - \prod_i (1 - \text{evidence}_{k,i})$$

4. TRAINING AND HYPERPARAMETERS

We initially considered fine-tuning ViT. With only 192 training images, we instead froze the backbone and trained a small head for a more repeatable baseline. Selected model: **Frozen ViT + logistic head**. Test data stayed outside fitting, model selection, and calibration.

Split protocol	192 train; 48 validation; 48 final test
Feature run	768-D; batch 8; cpu; 8.298s
Head	StandardScaler + multinomial logistic regression
Head settings	C=0.1; lbfgs; balanced; max_iter=5000
Frozen backbone	85.80M parameters; 0 trainable
Artifact thresholds	firearms 0.9914; explosives 0.1728; financial promotion 0.9946
Calibration	validation recall >=0.90; 12 positives per class
Head fit	train only; 0.0086s
Policy	1.0.0-pilot; review 0.35; margin 0.12

5. EVALUATION RESULTS

Final grouped test: n=48 (12 per class). Macro AP=1.000; group-bootstrap macro F1 95% interval=0.937 to 1.000 (2,000 source-group replicates).

TEST ACCURACY	MACRO F1
97.9%	97.9%
47 / 48 correct	target >0.93: met
ARGMAX RESTRICTED MEAN	THRESHOLD-OP MEAN
97.2%	94.4%
multiclass class-average	validation-selected thresholds
SAFE-CLASS PRECISION	SAFE-AD FALSE FLAG
1.000	0.0%
target >0.98: met	1 - Safe recall

Macro F1 met our >0.93 target. Safe precision met our >0.98 target; the Safe-ad false-flag rate was 0.0% (1 - Safe recall). Argmax restricted recall averaged 97.2%, but that is not the deployed threshold rule. At validation-selected thresholds, test recall was 91.7% for firearms, 91.7% for explosives, and 100.0% for financial promotion; mean 94.4%. The per-class >0.95 target fails for firearms and explosives.

Class	P	R	F1	n
Safe	1.000	1.000	1.000	12
Firearms	0.923	1.000	0.960	12
Explosives	1.000	0.917	0.957	12
Financial promo	1.000	1.000	1.000	12

Untouched test confusion matrix (counts)

True / pred	Safe	Fire	Expl	Fin
Safe	12	0	0	0
Firearms	0	12	0	0
Explosives	0	1	11	0
Financial promo	0	0	0	12

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6. BASELINE AND BACKBONE COMPARISON

We compared the selected ViT head with a ResNet-50 baseline. Validation and test results stay in separate columns [5, 6].

Model	Val F1	Test F1	Test acc	CPU p95
ViT primary	0.958	0.979	0.979	71.5 ms
ResNet50 baseline	0.957	0.873	0.875	40.1 ms

7. MEASURED LATENCY

MEDIAN	CLASSIFIER-PATH P95
69.1 ms	71.5 ms
batch 1	batch 1; excludes auxiliaries
MEAN	SERIAL THROUGHPUT
68.7 ms	14.6 img/s
single thread	derived; not concurrent

ViT classifier-path p95 was 71.5 ms, above our 50 ms latency budget. CPU protocol: 3 warmups, 20 runs, one thread. Full end-to-end p95 was not measured because file decoding, OCR, object detection, network transfer, application rendering are excluded. Concurrent throughput and MPS resource use were also not measured.

8. ENGINEERING CHALLENGES

Source style is tied to class labels.

We grouped campaigns before splitting and recomputed all 288 hashes. This controls direct leakage, but it cannot remove visual source cues.

Tuning could leak into the final test.

We fit the head on training data, selected restricted thresholds on validation, and recorded one post-calibration test protocol.

One grenade was misclassified as a firearm.

The saved failure was explosives classified as firearms (1/48 test images). We kept it in the audit table, and the saved policy result was REVIEW rather than automatic enforcement.

ViT exceeded the latency budget.

ViT classifier-path CPU p95 was 71.5 ms, above 50 ms, versus 40.1 ms for ResNet50. Neither value is end-to-end latency.

EXTERNAL DIAGNOSTIC

n=26; manually reviewed external Wikimedia diagnostic spot check; not a formal test set. Detector image recall was 100.0%, but the non-weapon detector false-positive rate was 90.0%; this is a generalization warning, not formal test evidence. The repository retains the complete case-level diagnostic record.

CLASSIFIER ACCURACY	MACRO F1
65.4%	55.5%
DETECTOR NONWEAPON FP	WEAPON BLOCK
90.0%	43.8%
SAFE NON-APPROVE	FINANCE CORRECT REVIEW
42.9%	33.3%

9. LIMITATIONS AND DEPLOYMENT BOUNDARY

- Class labels are tied to source style; the 48-image test has only 12 images per class and does not establish production performance.
- Threshold calibration used only 12 validation positives per class, so the selected operating points remain uncertain.
- The 216 third-party formal images remain local-only pending rights clearance; the 72 project-generated finance images carry no reuse license.
- Four single-label image categories omit mixed-policy ads, landing pages, advertiser identity, geography, and legal context.
- Occlusion sensitivity is a diagnostic, not a causal explanation; the external detector check has no box-level ground truth or mAP.
- Full-pipeline latency, concurrent throughput, and MPS resource use remain unmeasured; a human reviewer must make the final decision.

10. REPRODUCIBILITY

Run record	2026-08-25 random seed 462
Environment	Apple Silicon macOS CPU feature extraction
Data checks	Recomputed 288 image hashes; no split overlap
Model reload	Maximum prediction difference: 0

All performance figures were read from the saved evaluation records; dataset composition was checked against the formal registry. All 27 Wikimedia candidates were excluded from formal train, validation, and test evidence; 26 were retained for external diagnosis and one was rejected.

The repository contains the project specification, dataset registry, evaluation metrics, thresholds, model comparison, latency record, external diagnostic, and dataset-rights documentation.

The app's session charts describe only the cases run in that session. They are interface summaries, not production monitoring, throughput, or population evidence.

11. REFERENCES

- [1] [Google Ads policy overview](#)
- [2] [Interactive Advertising Bureau, Ad Product Taxonomy](#)
- [3] [Dosovitskiy et al., An Image is Worth 16x16 Words](#)
- [4] [Liu et al., Grounding DINO](#)
- [5] [ViT-B/16 AugReg model card](#)
- [6] [ResNet-50 A1 model card](#)
- [7] [ADautoGen-DS dataset card](#)
- [8] [Weapons Set 1 dataset card](#)
- [9] [Tesseract OCR repository](#)
- [10] [Wikimedia Commons licensing guidance](#)